

ARTIFICIAL INTELLIGENCE

Assessment Tool 1 – Analytical Problem Solving

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Question 3: Exam Timetabling

Constraint Satisfaction Problem (CSP)

Variables: AI, DBMS, OS, CN, DAA and SE. Each variable represents an exam.

Domains: Each exam can be assigned to T1, T2 or T3.

Constraints: $AI \neq DBMS$, $AI \neq OS$, $DBMS \neq CN$, $OS \neq CN$, $OS \neq DAA$, $CN \neq SE$, $DAA \neq SE$.

Goal: Assign a time slot to every exam without placing conflicting exams together.

Constraint Graph

An edge joins exams that cannot share a time slot.

DBMS

AI

CN

OS

SE

DAA

Q3: MRV + Backtracking Solution

One valid schedule (MRV tie-breaks may produce other valid schedules)

Choose AI = T1.

DBMS conflicts with AI, so choose DBMS = T2.

OS conflicts with AI and DBMS is not a direct conflict; choose OS = T2.

CN conflicts with DBMS and OS, so choose CN = T1.

DAA conflicts with OS, so choose DAA = T1.

SE conflicts with CN and DAA, so choose SE = T2.

Forward checking removes the selected time slot from each neighbouring exam's domain.

Q3: Forward Checking and AC-3

Reducing search

Forward checking: After assigning an exam, remove that time slot from all unassigned neighbouring exams.

If any unassigned exam has an empty domain, backtrack immediately.

AC-3 checks each constraint arc (X, Y) and removes values from X that have no compatible value in Y .

For simple inequality constraints, AC-3 may not prune much before assignments, but it detects domain wipe-outs early after domains become restricted.

The final assignment must satisfy all seven conflict constraints.

Question 4: Route Finding with A*

A* evaluates $f(n) = g(n) + h(n)$

Edges: S–A=2, S–B=5, A–C=4, A–D=7, B–D=3, C–D=1, C–G=6, D–G=2.

Heuristics: $h(S)=7$, $h(A)=6$, $h(B)=4$, $h(C)=3$, $h(D)=2$, $h(G)=0$.

$g(n)$ is the cost from S; $h(n)$ estimates the remaining cost; $f(n)$ is their sum.

Start at S with $g=0$ and $f=7$.

Q4: A* Open and Closed Lists

Open list ordered by lowest f-value; entries shown as node (g, h, f)

Initial: OPEN = [S(0,7,7)]; CLOSED = [].

Expand S: add A(2,6,8), B(5,4,9). CLOSED = [S].

Expand A: add C(6,3,9), D(9,2,11). CLOSED = [S,A].

Expand B: improve D to g=8, so D(8,2,10). CLOSED = [S,A,B].

Expand C: improve D to g=7, so D(7,2,9); add G(12,0,12). CLOSED = [S,A,B,C].

Expand D: improve G to g=9, so G(9,0,9). CLOSED = [S,A,B,C,D].

Select G: terminate. Path = $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$; total cost = 9.

Q4: Heuristic Checks and Greedy Search

Admissibility, consistency and comparison

True shortest costs to G: $h^*(G)=0$, $h^*(D)=2$, $h^*(C)=3$, $h^*(B)=5$, $h^*(A)=7$, $h^*(S)=9$.

Admissible: yes. Each supplied $h(n)$ is no greater than the true remaining cost.

Consistent: no. For edge A–C, $h(A)=6 > 4 + h(C)=7$ is false (so this edge is okay); for S–A, $7 \leq 2+6=8$; for A–D, $6 \leq 7+2=9$. All edges satisfy consistency, including C–G: $3 \leq 6$. Thus it is consistent.

Greedy Best-First uses h only: it chooses B ($h=4$) over A ($h=6$), then D ($h=2$), then G ($h=0$). Path $S \rightarrow B \rightarrow D \rightarrow G$ costs 10.

A* finds cost 9, while Greedy finds cost 10 for this graph.

Question 5: 8-Puzzle

Problem formulation

State: arrangement of the eight numbered tiles and one blank in a 3×3 board.

Initial state: 2 8 3 / 1 6 4 / 7 _ 5.

Goal state: 1 2 3 / 8 _ 4 / 7 6 5.

Actions: move the blank up, down, left or right when the move stays inside the board.

Transition model: swap the blank with the adjacent tile in the chosen direction.

Path cost: 1 per move; the objective is to reach the goal with minimum moves.

Q5: Heuristics at the Start

Blank is not counted as a tile

$h_{\blacksquare}$ (misplaced tiles): count tiles not in their goal positions.

Misplaced tiles are 2, 8, 1 and 6, so $h_{\blacksquare} = 4$.

$h_{\blacksquare}$ (Manhattan distance): sum $|\text{current row} - \text{goal row}| + |\text{current column} - \text{goal column}|$ for every tile.

Distances: tile 2 = 1; 8 = 2; 1 = 1; 6 = 1; tiles 3, 4, 5 and 7 = 0.

Therefore $h_{\blacksquare} = 1 + 2 + 1 + 1 = 5$.

Q5: A* Solution Path

Using Manhattan distance h ; each move costs 1

Start: 2 8 3 / 1 6 4 / 7 _ 5

Move blank UP: 2 8 3 / 1 _ 4 / 7 6 5

Move blank LEFT: 2 8 3 / _ 1 4 / 7 6 5

Move blank UP: _ 8 3 / 2 1 4 / 7 6 5

Move blank RIGHT: 8 _ 3 / 2 1 4 / 7 6 5

Move blank DOWN: 8 1 3 / 2 _ 4 / 7 6 5

Move blank LEFT: 8 1 3 / _ 2 4 / 7 6 5

Move blank UP: _ 1 3 / 8 2 4 / 7 6 5

Move blank RIGHT: 1 _ 3 / 8 2 4 / 7 6 5

Move blank DOWN: 1 2 3 / 8 _ 4 / 7 6 5 (GOAL). Total = 9 moves.

Q5: Search Tree and Heuristic Dominance

Illustrative solution branch in the A* search tree

Start ($h=5$)

■ ■ U: 2 8 3 / 1 _ 4 / 7 6 5 ($h=4$)

■ ■ L: 2 8 3 / _ 1 4 / 7 6 5 ($h=5$)

■ ■ U: _ 8 3 / 2 1 4 / 7 6 5 ($h=6$)

■ ■ R → D → L → U → R → D → Goal

h dominates h because Manhattan distance is at least as informed as misplaced-tile count: each misplaced tile must move at least one step.

Note: the branch is a compact illustration of the solution path, not every node expanded by A*.

Question 9: Smart Traffic Signal Controller

Intelligent agent design

Most suitable architecture: utility-based agent with a model-based component.

It uses camera counts and emergency-vehicle detection to choose signal phases and green durations.

Utility goal: reduce average waiting time, queue length and congestion while giving emergency vehicles priority.

A model-based component tracks current signal phase, vehicle queues and recent observations.

A learning component could improve timing from historical traffic data, but it is not essential to the basic design.

Q9: Agent Architecture

Perceive → update state → evaluate utility → act

Sensors

Cameras / emergency detector

State / Model

Queues, phase, timers

Utility Evaluation

Wait time + priority

Action Selector

Actuators

Traffic lights

Feedback: updated camera observations are sensed again after each signal change.

Q9: Decision-Cycle Pseudocode

Simple pseudocode

LOOP continuously:

percept $\leftarrow$ read camera vehicle counts and detect emergency vehicles

state $\leftarrow$ update queues, current phase and elapsed green time

IF an emergency vehicle is detected:

select a safe priority phase for its approach

ELSE:

estimate waiting time for each approach

choose the phase with the highest utility (lower wait/queue is better)

set green duration based on queue size and safety limits

change signal safely; repeat

Question 12: Mars Rover

Intelligent agent for sample analysis and exploration

Objective: explore Mars, identify scientifically useful rocks/soil, collect samples and send data to Earth.
The rover must operate with limited energy, communication delays and uncertain terrain.
Its decisions should balance scientific value, safety, time and power consumption.

Q12: Percepts and Environment

What the rover senses and where it operates

Percepts: camera images; distance/obstacle readings; wheel and position data; terrain slope; temperature and power level; spectrometer and other instrument readings; sample properties; communication status.

Environment: Mars surface, rocks, soil, slopes, dust, changing light and temperature, and possible hazards.

Partially observable: sensors provide incomplete information about the surroundings.

Stochastic/uncertain: terrain and wheel interactions may not behave exactly as expected.

Sequential and dynamic: each movement affects later choices, and environmental conditions can change.

Mostly continuous, single rover agent interacting with a physical environment; Earth operators may also provide commands.

Q12: Actions, Performance and Architecture

Design choices

Actions: drive/turn; stop; avoid obstacles; take images; use robotic arm; drill or scoop soil/rock; place samples for analysis; run scientific instruments; store and transmit data; manage power; enter safe mode.

Performance measures: useful samples and discoveries; quality/amount of scientific data; distance explored; successful sample analysis; safe operation; energy efficiency; mission objectives completed; reliable communication.

Suitable architecture: model-based, goal-based and utility-based hybrid agent.

Why: it maintains an internal map/state, plans toward exploration and sampling goals, and chooses actions by balancing scientific benefit against risk, energy and time.

Learning can be added to improve terrain classification and route choices from experience.

Conclusion

Answers based on Assessment Tool 1

Q3: CSP formulation, conflict graph, MRV/backtracking and forward checking.

Q4: A* gives shortest route $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$ with cost 9; Greedy route costs 10.

Q5: 8-puzzle formulation, $h_{min}=4$, $h_{max}=5$, and a 9-move solution path.

Q9: Utility-based traffic agent with perception, state update, utility evaluation and signal control.

Q12: Mars Rover senses terrain and samples, performs safe scientific actions and uses a hybrid goal/utility-based architecture.